STAT452: Variable Selection and Regularization — Lesson 3 Notes

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1 Motivation for Variable Selection

1.1 Why Perform Variable Selection?

- **Simplicity (Occam's Razor):** The simplest adequate model is best. Redundant predictors should be removed to maintain interpretability.
- **Noise Reduction:** Unnecessary predictors add noise and obscure true relationships, wasting degrees of freedom and reducing estimation precision.
- **Collinearity Mitigation:** Too many variables may attempt to explain the same effect, leading to unstable and inflated coefficient estimates.
- **Cost Minimization:** Fewer predictors mean lower data collection costs and faster models, which is especially important in predictive modeling.

1.2 Preparation Before Selection

- Screen for **outliers and influential points**; consider removing or treating them temporarily.
- Apply **appropriate transformations** (normalization, log-transform, standardization) as needed.

2 The Problem with p-Value-Based Selection

2.1 The Wald Test Statistic

To test $H_0 : \beta_i = 0$, we use the Wald test statistic:

$$t = \frac{\hat{\beta}_i}{\mathrm{SE}(\hat{\beta}_i)}$$

Under H_0, this follows a $t_{n-(p+1)}$ distribution and tends to the standard normal as $n \to \infty$.

2.2 Decomposition of the Test Statistic

The test statistic can be decomposed as:

$$t = \frac{\hat{\beta}_i}{\dfrac{\hat{\sigma}}{\sqrt{n \cdot \widehat{\mathrm{Var}}(X_i)}} \cdot \sqrt{\mathrm{VIF}_i}}$$

The denominator (standard error) depends on:

1. **Noise level ($\hat{\sigma}$):** More noise inflates the standard error $\Rightarrow$ lower significance.

2. **Sample size (n):** Larger n shrinks the standard error ($\sim 1/\sqrt{n}$) $\Rightarrow$ higher significance.
3. **Predictor variance ($\widehat{\mathbf{Var}}(X_i)$):** More spread in X_i improves estimation $\Rightarrow$ higher significance.
4. **Collinearity ($\mathbf{VIF}_i$):** High multicollinearity inflates the standard error $\Rightarrow$ lower significance.

2.3 Why p-Values Are Misleading for Selection

- The test statistic reflects **data quality and predictor structure**, not just effect size.
- **Unimportant** variables with high variance or low multicollinearity may appear significant.
- **Important** variables may appear non-significant if measured with high uncertainty.
- As $n \to \infty$, even **tiny nonzero coefficients** will become statistically significant.
- p-value-based selection favors predictors with high variance, low correlation to others (low VIF), and large sample sizes — regardless of practical relevance.

Better alternatives: Cross-validation (CV), Information criteria (AIC, BIC), Regularization methods (LASSO, Ridge).

3 Cross-Validation for Variable Selection

Cross-validation estimates the model's ability to predict **unseen data**. While C_p, AIC, and BIC attempt this analytically, CV does it empirically.

3.1 Leave-One-Out Cross-Validation (LOOCV)

- For each observation $i = 1, \dots, n$: fit the model on all data *except* observation i, then predict $\hat{y}_{(i)}$.
- The LOOCV error is:
$$\text{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_{(i)})^2$$
- **Properties:** Low bias (nearly the full dataset is used for training), but higher variance and computationally expensive (n models to fit).
- Tends to **overfit** (includes unnecessary variables).

3.2 k-Fold Cross-Validation

- Randomly partition the data into k roughly equal-sized folds (typically $k = 5$ or 10).
- For each fold j: train on the remaining $k - 1$ folds, predict on fold j, and compute MSE_j.
- The k-fold CV error is:
$$\text{CV}_{(k)} = \frac{1}{k}\sum_{j=1}^{k}\text{MSE}_j$$
- **Properties:** More computationally efficient (only k models to train), less variance in error estimates, slightly less accurate if all models are misspecified.

4 Information Criteria

4.1 Akaike Information Criterion (AIC)

- **Definition:**
$$\text{AIC} = -2\log\hat{L} + 2p$$
where $\hat{L}$ is the maximum likelihood of the fitted model and p is the number of parameters (including intercept).
- **Interpretation:** AIC estimates the relative out-of-sample prediction error. It penalizes model complexity to avoid overfitting. **Lower AIC values** indicate a better trade-off between fit and simplicity.
- **Use when:** Your goal is **prediction**, not necessarily finding the "true" model.

4.2 Bayesian Information Criterion (BIC)

- **Definition:**
$$\text{BIC} = -2\log\hat{L} + p\log n$$
where n is the sample size.
- **Interpretation:** BIC penalizes complexity **more strongly** than AIC (because of the $\log n$ term). As $n \to \infty$, BIC tends to select the correct (true) model if it exists. **Lower BIC is better.**
- **Use when:** You prioritize **simpler models** and model identification over pure predictive performance.

4.3 Mallows' C_p Criterion

- **Definition:**

$$C_p = \frac{\mathrm{RSS}_p}{\hat{\sigma}^2} - n + 2p$$

  where RSS_p is the Residual Sum of Squares from a model with p predictors, $\hat{\sigma}^2$ is the error variance estimate from the **full model** (with all predictors), and n is the number of observations.

- **Interpretation:** If the subset model is unbiased, then $C_p \approx p$. Models with $C_p \approx p$ and small overall value are preferred.

- **Use when:** You trust the full model as a benchmark to estimate $\hat{\sigma}^2$.

4.4 Comparison: AIC vs. BIC vs. C_p

Criterion	**Penalty Strength**	**Best Use Case**
AIC	Lenient ($2p$)	When **prediction** is the primary goal.
BIC	Stronger ($p \log n$)	When identifying the "true" model or a **simpler model** is preferred.
C_p	Error-based	When the **full model** variance estimate $\hat{\sigma}^2$ is trustworthy.

5 Stepwise Selection Methods

Stepwise methods add or remove one variable at a time based on a selection criterion (AIC, BIC, Adjusted R^2, Mallows' C_p, CV error, etc.).

5.1 Forward Stepwise Selection

1. Start with **no predictors** (intercept-only model).
2. At each step, add the variable that **most improves** the model (e.g., lowest AIC).
3. Repeat until no further significant improvement is obtained.

5.2 Backward Stepwise Selection

1. Start with the **full model** (all predictors included).
2. At each step, remove the variable whose exclusion **improves performance most**.
3. Continue until further removal worsens the model.

5.3 Mixed (Forward-Backward) Stepwise Regression

1. Start with an initial model (e.g., intercept-only).
2. Add the predictor that most improves the model (e.g., lowest AIC or highest F-statistic).
3. After adding, **re-evaluate all included variables**.
4. Remove any variable that no longer contributes significantly.
5. Repeat steps 2–4 until no further change is possible.

Key point: Variables already added may still be removed later — this is the distinguishing feature from pure forward selection.

5.4 Stepwise vs. All-Subsets Selection

- Stepwise methods only explore a **subset of possible models** and are computationally efficient but may miss the globally best model.
- All-subsets selection considers all 2^p models (feasible only for small p).
- Stepwise methods are a **compromise**: faster but potentially suboptimal.

6 Ridge Regression (L2 Regularization)

6.1 Limitations of OLS Motivating Regularization

- **Multicollinearity:** When predictors are highly correlated, $\mathbf{X}^T\mathbf{X}$ becomes nearly singular, producing unstable and highly sensitive coefficient estimates.
- **High Variance:** As p approaches or exceeds n, OLS predictions become highly variable with poor generalization.
- **No Variable Selection:** OLS includes all predictors regardless of relevance, reducing interpretability.

6.2 Ridge Regression Formulation

Ridge regression solves the following penalized optimization problem:

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}$$

Equivalent constrained form:

$$\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 \leq t$$

Matrix notation and closed-form solution:

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2 \right\} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

Key properties:

- Adding $\lambda\mathbf{I}$ to $\mathbf{X}^T\mathbf{X}$ ensures invertibility even under multicollinearity.
- The intercept β_0 is **not penalized**.
- Predictors should be **standardized** before fitting to ensure fair penalization across variables.

6.3 Standardization and Intercept Treatment

Before estimating Ridge coefficients:

- **Intercept:** Set $\hat{\beta}_0 = \bar{y}$; the penalty applies only to slope coefficients $(\beta_1, \beta_2, \ldots, \beta_p)$.
- **Standardization:** Transform predictors to ensure comparability:

  $$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{k=1}^{n}(x_{kj} - \bar{x}_j)^2}}$$

  This prevents penalization from disproportionately affecting variables with larger scales.

6.4 The Bias-Variance Trade-Off

- The Mean Squared Error decomposes as: $\text{MSE} = \text{Bias}^2 + \text{Variance}$.
- Increasing model complexity reduces bias but increases variance.
- Ridge slightly **increases bias** but **significantly reduces variance**, finding a favorable balance.

6.5 Bias of Ridge Estimates

- **OLS estimate:** $\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$
- **Ridge estimate:** $\hat{\boldsymbol{\beta}}_{\text{ridge}} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}^T\mathbf{Y}$
- **Expected value:**
$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{ridge}}] = (\mathbf{I}_p + \lambda(\mathbf{X}^T\mathbf{X})^{-1})^{-1}\boldsymbol{\beta} \neq \boldsymbol{\beta}$$
$\Rightarrow$ Ridge estimates are **biased**.
- If predictors are standardized and uncorrelated:
$$\hat{\beta}_{j,\text{ridge}} = \frac{1}{1+\lambda}\hat{\beta}_{j,\text{OLS}}$$
- Variance of Ridge estimates is **smaller** than that of OLS, especially when data has multicollinearity.

6.6 The Tuning Parameter λ

- $\lambda = 0$: Ridge reduces to OLS.
- $\lambda \to \infty$: All coefficients shrink to zero.
- λ is chosen using **cross-validation** to minimize prediction error.

6.7 Effective Degrees of Freedom

In Ridge regression, the degrees of freedom is no longer simply p:

$$\text{df}(\lambda) = \text{tr}\left[(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{X}\right] = \sum_{j=1}^{p} \frac{d_j^2}{d_j^2 + \lambda}$$

where d_j are the singular values of $\mathbf{X}$. As $\lambda \to \infty$, $\text{df}(\lambda) \to 0$.

6.8 Why Ridge Keeps All Variables

- The L_2 penalty discourages large coefficients by shrinking them towards zero, but **never exactly to zero**.
- **Geometric intuition:** The constraint region for Ridge is a **ball** (circle/sphere). RSS contour ellipses typically touch the ball boundary at non-zero values.
- Hence, all predictors are **retained** in the model with reduced magnitudes.

7 LASSO Regression (L1 Regularization)

7.1 LASSO Formulation

LASSO (Least Absolute Shrinkage and Selection Operator) minimizes:

$$\hat{\boldsymbol{\beta}}_{\text{lasso}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

Equivalent constrained form:

$$\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \leq t$$

Matrix notation:

$$\hat{\boldsymbol{\beta}}_{\text{lasso}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1 \right\}$$

7.2 Properties of LASSO Estimates

- **No closed-form solution:** The L_1 penalty involves absolute values, which are non-differentiable at zero. Solutions are nonlinear in y_i due to the non-smooth constraint.
- **Biased** (towards zero), but with **smaller variance** than OLS.
- Does **not perform as well** as Ridge when data have strong multicollinearity problems (may unstably select only one variable from a correlated group).
- **Greatest advantage — Sparsity:** LASSO can set some coefficients **exactly equal to zero** when λ is sufficiently large, performing **automatic variable selection**.

7.3 Sparsity: The Key Advantage

In a model $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$, we may believe many of the β_j's are actually zero. LASSO seeks a set of **sparse solutions**:

- **Geometric intuition:** The constraint region for LASSO is a **diamond** (in 2D) or a cross-polytope. RSS contour ellipses are more likely to touch the diamond at a **corner** (axis), where one or more coefficients are exactly zero.
- This means LASSO performs **model selection** automatically.

7.4 Choosing λ via Cross-Validation

1. Try a range of λ values.
2. For each λ: use the training set to train the model, predict values on the test/validation set.
3. Compute the Root Mean Square Error:

$$\text{RMSE} = \sqrt{\frac{\sum_{\text{test}} (y_i - \hat{y}_i)^2}{n_{\text{test}}}}$$

4. Choose the λ that has the **smallest RMSE** (or use k-fold CV and average).

8 Comparison: OLS vs. Ridge vs. LASSO

Property	**OLS**	**Ridge (L2)**	**LASSO (L1)**
Objective	$\min \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$	$+\lambda \sum \beta_j^2$	$+\lambda \sum \|\beta_j\|$
Constraint shape	None	Ball (circle)	Diamond
Bias	Unbiased	Biased	Biased
Variance	Highest	Reduced	Reduced
Variable selection	No	No (all kept)	Yes (sparsity)
Closed-form	Yes	Yes	No
Best when	$p \ll n$, low collinearity	All predictors relevant, multicollinearity	Sparse true model